

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Code Layout, Readability & Reusability	DATE	July 2026

Code Layout, Readability & Reusability

Project Structure

- credit_card_approval.ipynb - data analysis and model training notebook
- app.py - Flask backend
- templates/home.html, index.html, result.html - frontend pages
- model.pkl - serialized trained model

Readability Practices

- Descriptive variable names throughout (e.g., X_train, y_test, rf_model)
- Model training logic wrapped in named functions (random_forest(), d_tree()) per project story requirements
- Clear separation between data preprocessing, model training/evaluation, and deployment code
- Encoding maps in app.py are named explicitly (e.g., GENDER_MAP, INCOME_TYPE_MAP) for clarity

Reusability

- The preprocessing pipeline can be re-applied to new applicant batches without modification
- Model training functions accept X_train/X_test/y_train/y_test as parameters, allowing reuse across different data splits
- The Flask app cleanly separates encoding logic from prediction logic, so the model can be swapped without changing the frontend